

A REPORT
ON
APEX PORTAL

Submitted by

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Under the guidance of,

Mr.Vishnu Shankar

Ms.Akkamahadevi

in partial fulfillment for the award of the

degree of

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE AND ENGINEERING
(COMPUTER ENGINEERING)**

AT



PRESIDENCY UNIVERSITY

BENGALURU

AUGUST 2026

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



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CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on “**APEX PORTAL**” being submitted by “**CHETHAN G LOKLLER**” bearing roll number “**20241CSE0577**” in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering (Computer Science and Engineering)** is a bonafide work carried out under our supervision.

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DECLARATION

I hereby declare that the work, which is being presented in the report entitled “**APEX PORTAL:CONSTRUCTION** ” in partial fulfillment for the award of Degree of **Bachelor of Technology in Computer Science and Engineering (Computer Science and Engineering)**, is a record of my own investigations carried under the guidance of **Dr.vishnu shankar, Assistant professor and Dr.Akkamahadevi, Assistant professor** , Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

Name:

Roll No:

Signature of the Student:

INTERNSHIP COMPLETION CERTIFICATE

➤ Add Internship/Course completion certificate issued from an organization. It must have the duration of the Internship, i.e. start and end date, Internship/RPL/Skill based course title and a technology on which work is carried out.

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ABSTRACT

APEX PORTAL – Construction Safety is an AI-powered construction-site safety monitoring platform developed to address common workplace safety challenges such as improper use of Personal Protective Equipment (PPE), unsafe working practices, and delayed identification of safety violations. Construction sites contain multiple hazards, and continuous manual monitoring of workers can be difficult, time-consuming, and less efficient. The proposed system uses **Artificial Intelligence (AI), Computer Vision, and Deep Learning** to automate the monitoring process and provide faster safety assessment.

The platform uses the **YOLOv11 (Ultralytics) object detection model** to analyze construction-site images and video streams and detect workers along with essential PPE such as **helmets and safety vests**. A dataset is prepared and managed using **Roboflow**, while model training and development are performed using **Google Colab with NVIDIA GPU acceleration**. **Python and OpenCV** are used for image and video processing, while **Pandas and Matplotlib** support data handling and visualization.

When a worker is detected without the required PPE, the system identifies the corresponding safety violation and records relevant information for further analysis. The collected data can be used to determine **PPE compliance, violation counts, safety scores, risk levels, and overall safety performance**. The platform also supports safety monitoring and analysis through an interactive **Streamlit dashboard**, allowing users to view safety information,

analytics, violation records, and reports in an organized manner.

The main goal of **APEX PORTAL – Construction Safety** is to reduce dependence on continuous manual inspection and provide an intelligent approach to construction-site safety management. By combining AI-based detection with data analysis and dashboard visualization, the platform enables faster identification of safety violations and supports better safety decision-making. Overall, the project demonstrates how modern AI and Computer Vision technologies can be applied to create a **more efficient, automated, and data-driven construction safety monitoring system.**

Introduction (Applicable to industry projects)

APEX PORTAL – Construction Safety

The construction industry involves various workplace hazards, making **worker safety and proper use of Personal Protective Equipment (PPE)** essential. Workers are generally required to use safety equipment such as **helmets, safety vests, and other protective gear** to reduce the risk of accidents and injuries. However, continuous manual monitoring of workers can be challenging, especially on large construction sites with many workers and ongoing activities.

With the advancement of **Artificial Intelligence (AI), Computer Vision, and Deep Learning**, automated safety monitoring can provide an effective solution. AI-based vision systems can analyze images and video footage to identify workers, monitor PPE usage, and detect safety violations more efficiently.

APEX PORTAL – Construction Safety is developed as an AI-based construction safety monitoring platform that uses **YOLOv11 object detection** to identify workers and monitor PPE compliance. The system can detect violations such as **missing helmets and safety vests** from construction-site images and videos. The detected information is processed and analyzed to provide safety insights such as compliance levels, violation records, safety scores, and risk assessment.

The platform integrates technologies such as **Python, YOLOv11, Roboflow, OpenCV, Google Colab, Pandas, Matplotlib, and Streamlit** to create an end-to-end safety monitoring solution. The Streamlit dashboard provides an interactive interface for viewing safety analytics, monitoring violations, and generating reports.

The main purpose of **APEX PORTAL – Construction Safety** is to support the construction industry with **automated, intelligent, and data-driven safety monitoring**. By reducing dependence on manual inspection and enabling faster identification of PPE

violations, the system can contribute to improved safety awareness, better compliance, and a safer construction working environment.

Objective

The main objective of **APEX PORTAL – Construction Safety** is to develop an AI-based platform that improves construction-site safety through automated monitoring and intelligent analysis.

The key objectives are:

1. **To develop an AI-powered construction safety monitoring system** using Artificial Intelligence and Computer Vision.
2. **To detect construction workers automatically** from images and video using the YOLOv11 object detection model.
3. **To monitor PPE compliance** by detecting safety equipment such as helmets and safety vests.
4. **To identify safety violations** such as missing helmets or safety vests and record the detected violations.
5. **To support real-time safety monitoring** and enable faster identification of unsafe conditions.

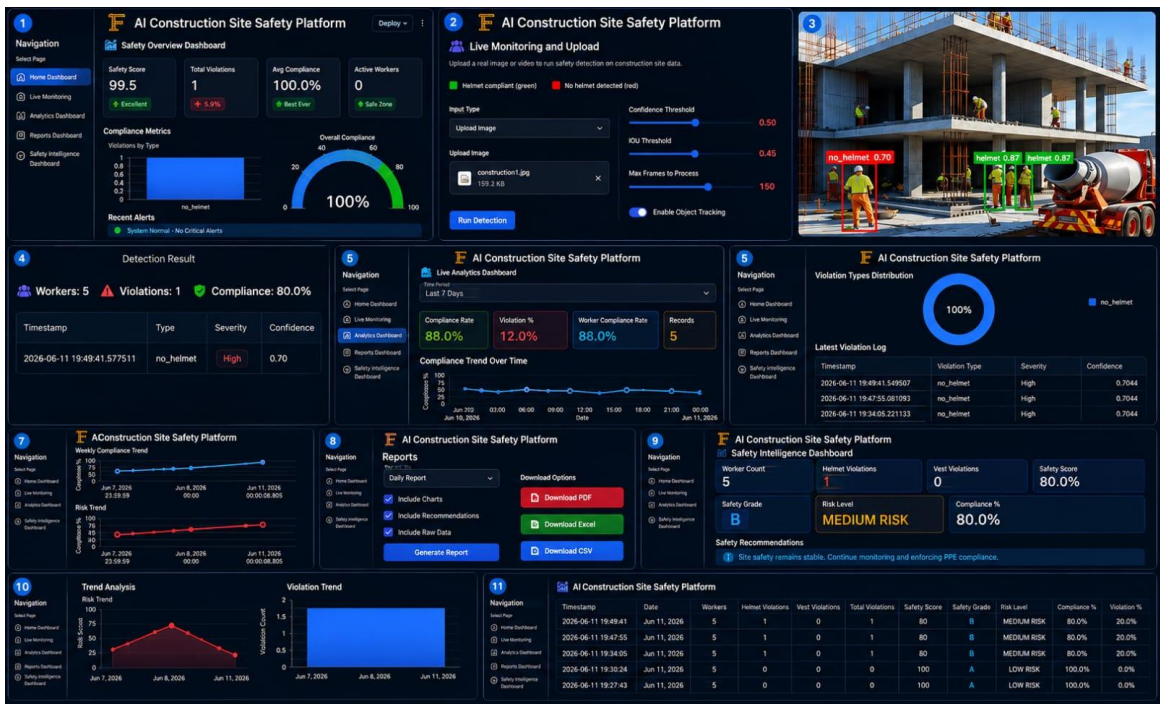
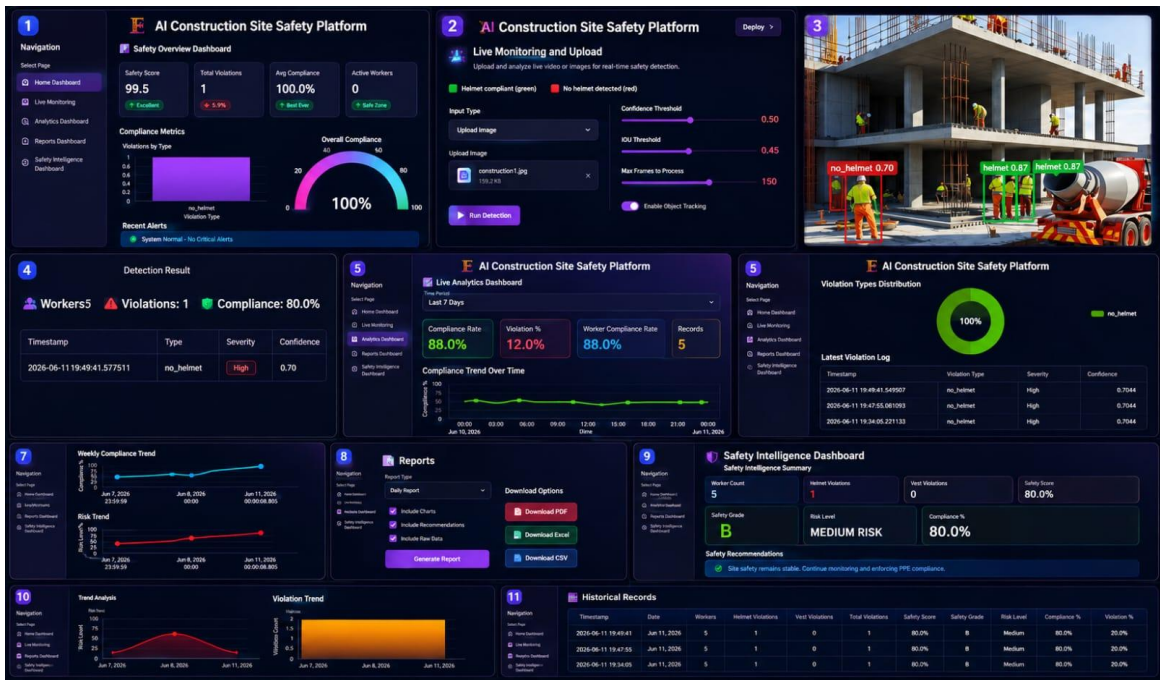
6. **To calculate safety indicators** such as PPE compliance percentage, safety score, violation count, and risk level.
7. **To provide an interactive dashboard** for displaying safety monitoring results, analytics, and reports.
8. **To reduce dependency on manual safety inspections** by automating routine PPE monitoring.
9. **To support data-driven safety decisions** through safety analytics and historical violation records.
10. **To improve overall construction-site safety** by enabling faster detection and response to safety risks.

Work progress

- **Requirement Analysis:** Identified construction-site safety challenges and defined the project requirements.
- **Dataset Preparation:** Collected and organized construction-site images and PPE data using **Roboflow**.
- **Data Annotation:** Prepared labeled datasets for workers, helmets, and safety vests.
- **Model Development:** Trained the **YOLOv11** object detection model using Google Colab and GPU acceleration.
- **PPE Detection:** Implemented detection of workers and PPE compliance, including helmets and safety vests.
- **Safety Violation Detection:** Developed functionality to identify missing PPE and record safety violations.
- **Image & Video Processing:** Integrated **OpenCV** for processing construction-site images and video.

- **Safety Analysis:** Implemented safety score, compliance percentage, violation tracking, and risk-level analysis.
- **Dashboard Development:** Developed an interactive **Streamlit dashboard** for monitoring and visualizing safety information.
- **Reporting:** Implemented safety analytics, historical records, and report generation.
- **Current Status:** Core AI detection, safety analysis, dashboard, and reporting features have been developed and integrated into **APEX PORTAL – Construction Safety**.

Results



Conclusion

APEX PORTAL – Construction Safety successfully demonstrates the use of **Artificial Intelligence, Computer Vision, and Deep Learning** for automated construction-site safety monitoring. The platform uses **YOLOv11** to detect workers and monitor PPE compliance, helping identify violations such as missing helmets and safety vests.

The system provides **safety analysis, compliance monitoring, risk assessment, dashboards, and reports** through an interactive platform. By reducing dependence on manual inspection and enabling faster identification of safety violations, **APEX PORTAL** can support better safety management, improve PPE compliance, and contribute to a **safer and more efficient construction environment**.

Reference

1. **Ultralytics YOLO Documentation** – YOLOv11 object detection and model implementation.
2. **Roboflow Documentation** – Dataset collection, annotation, and preparation for computer vision.
3. **OpenCV Documentation** – Image and video processing techniques.
4. **Python Documentation** – Programming and data-processing concepts.
5. **PyTorch Documentation** – Deep learning framework used for model development and execution.
6. **Streamlit Documentation** – Development of interactive web dashboards and data applications.
7. **Google Colab Documentation** – Cloud-based environment for AI model training and experimentation.
8. **Pandas Documentation** – Data processing and analysis.
9. **Matplotlib Documentation** – Data visualization and graphical analysis.
10. **Construction Safety and PPE Guidelines** – General principles of personal protective equipment and workplace safety monitoring.